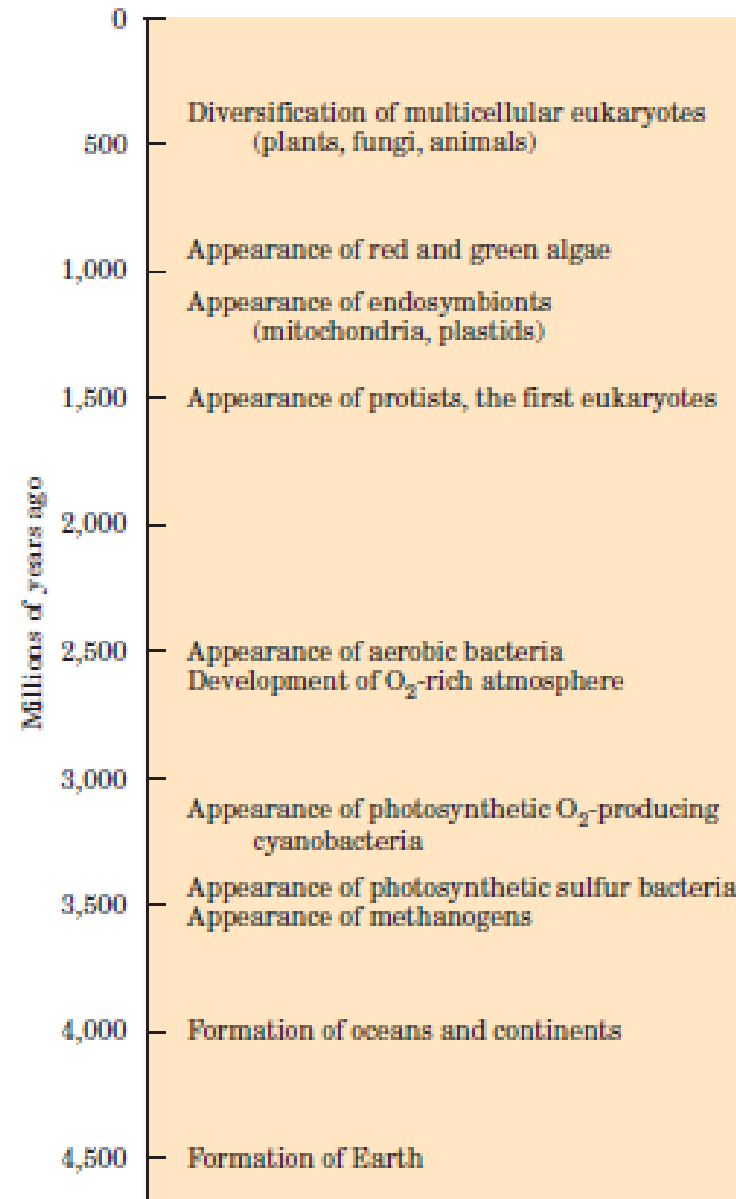




What is Life?

What are the principles of Biology ?

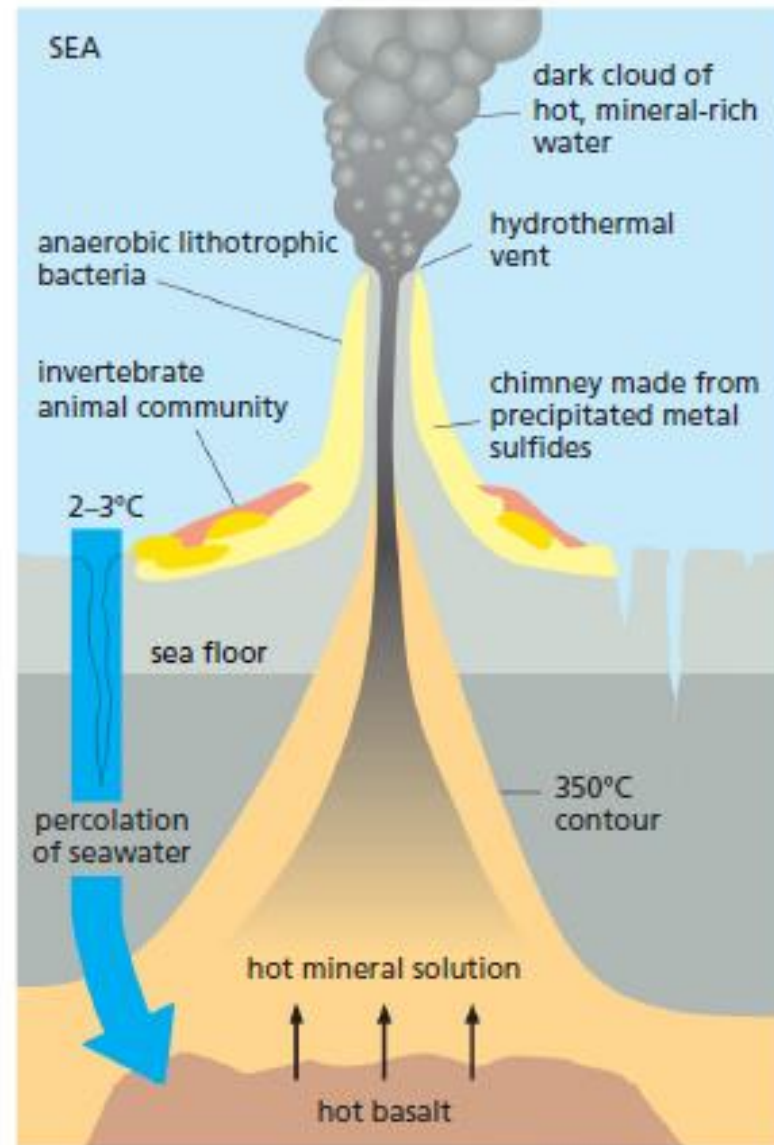
Landmarks in the evolution of life on Earth



Life's Origins

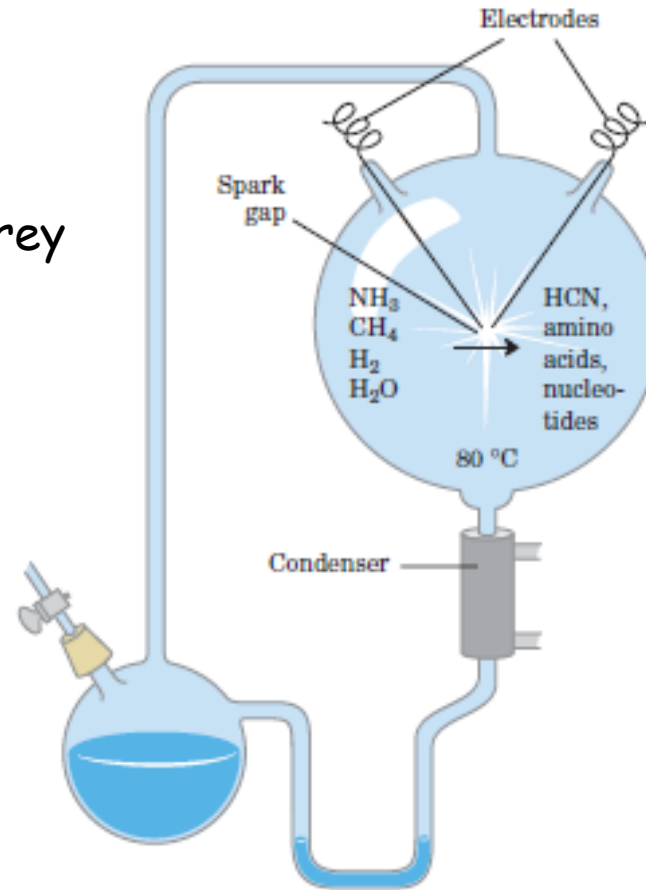
Today, research into the origin of life is interdisciplinary with workers trying to answer four main questions:

1. What was the Earth's physical environment like when life first evolved?
2. What sorts of chemical reactions could produce the building blocks of life and could these occur naturally in the early Earth's environment?
3. How could the complex organic molecules be compartmentalized into a contained unit?
4. How did the genetic code evolve?



How did the first living organisms acquire their characteristic organic building blocks?

Miller and Urey Experiment



Hypothesis:

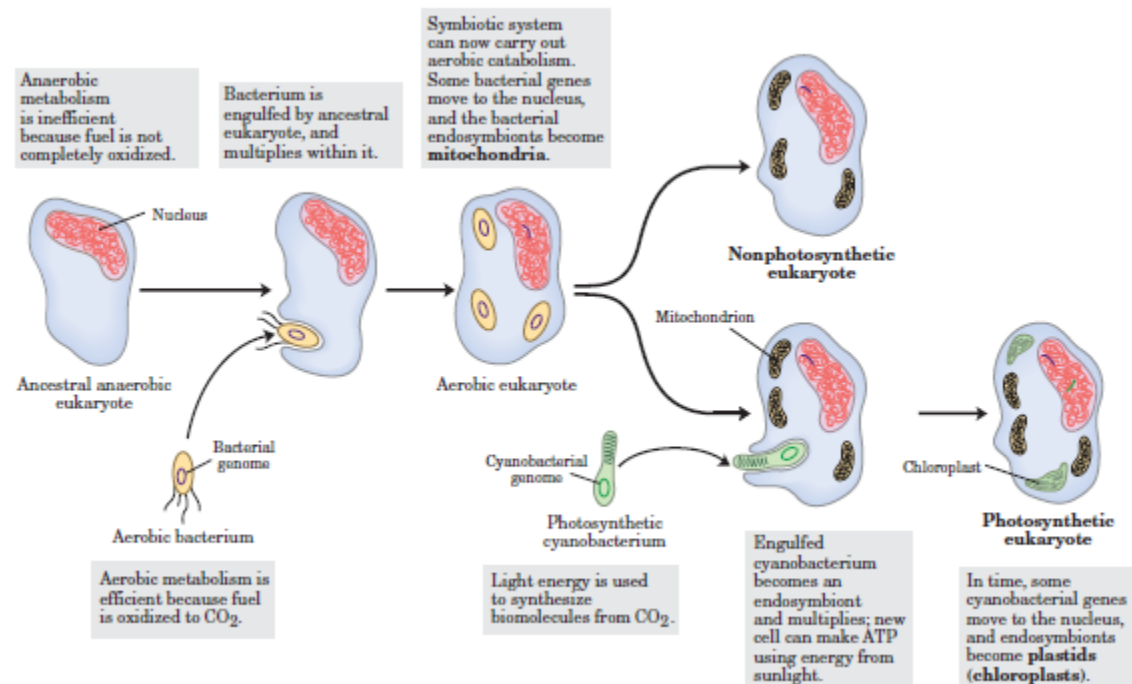
Effects of powerful atmospheric forces—ultraviolet irradiation, lightning, or volcanic eruptions—on the gases in the prebiotic Earth's atmosphere, and on inorganic solutes in superheated thermal vents deep in the ocean.

The appearance of the first living cell???

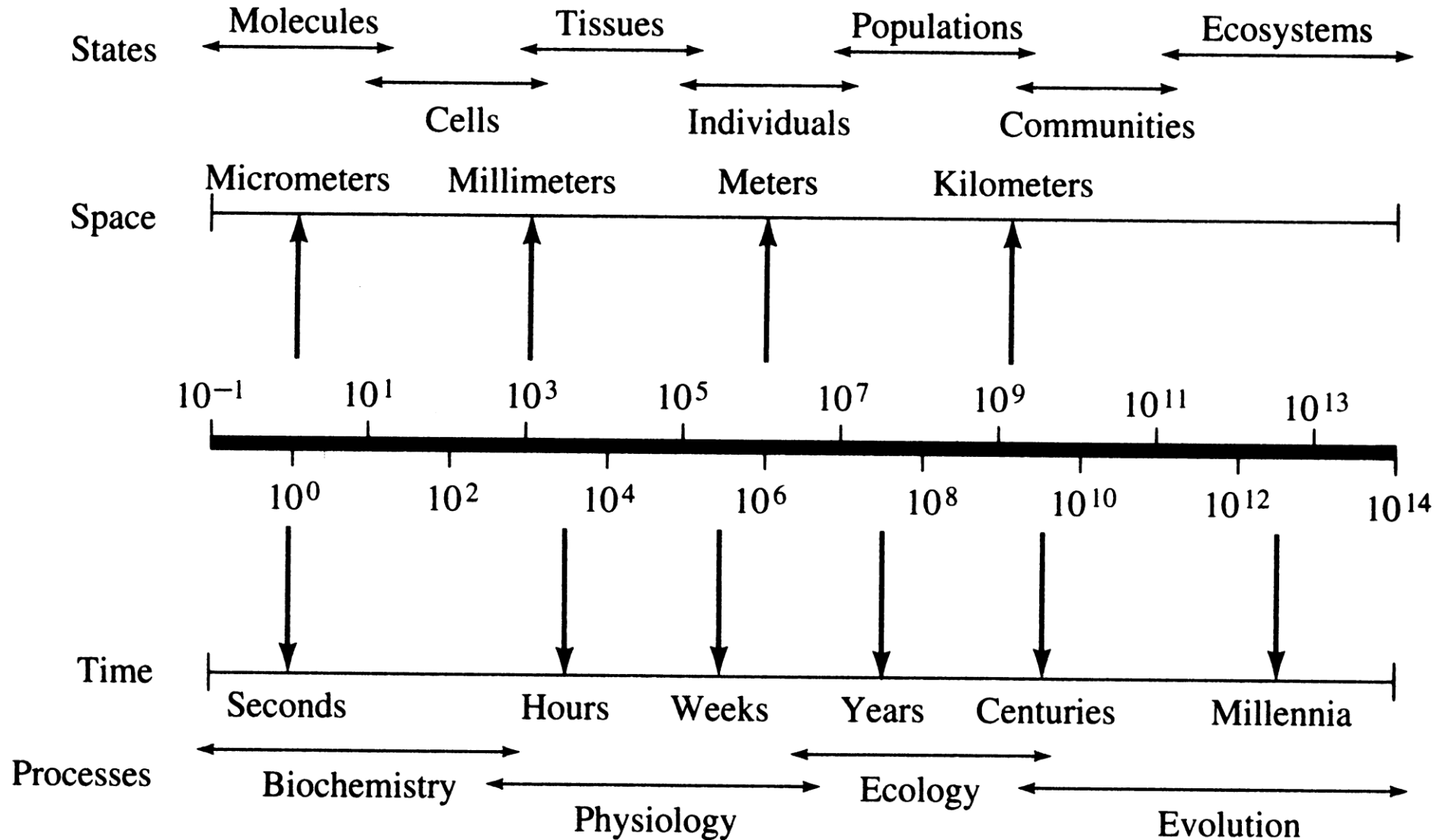
The First Cell Used Inorganic Fuels???

Lynn Margulis and the Theory of Endosymbiosis

- Eukaryotic cells originated from a series of endosymbiotic events involving multiple prokaryotes
- This idea was considered outrageous at the time (1967), but many of Margulis's ideas have since become widely accepted
- Mitochondria are the descendants of oxygen-respiring bacteria and chloroplasts were originally photosynthetic bacteria, are almost universally accepted



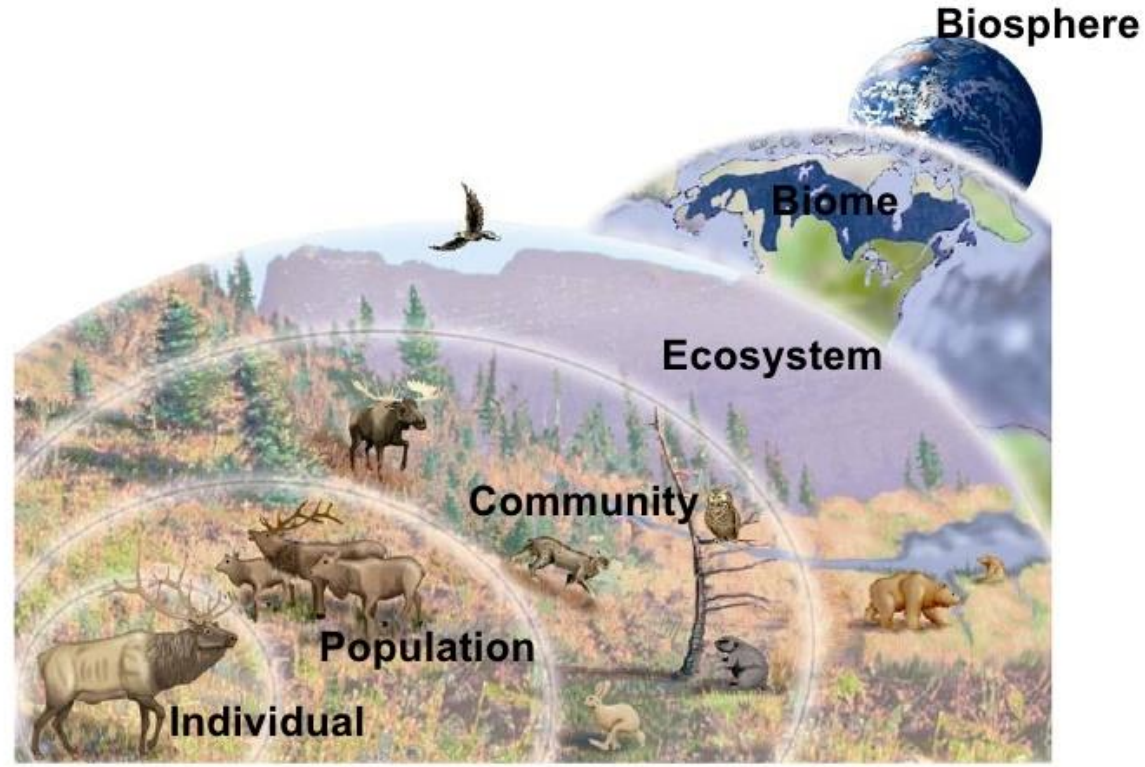
Biology is a vast discipline



The Hierarchical Structure of Life



Levels of Organization



Individual = 1 Species

Population = Many of the same species

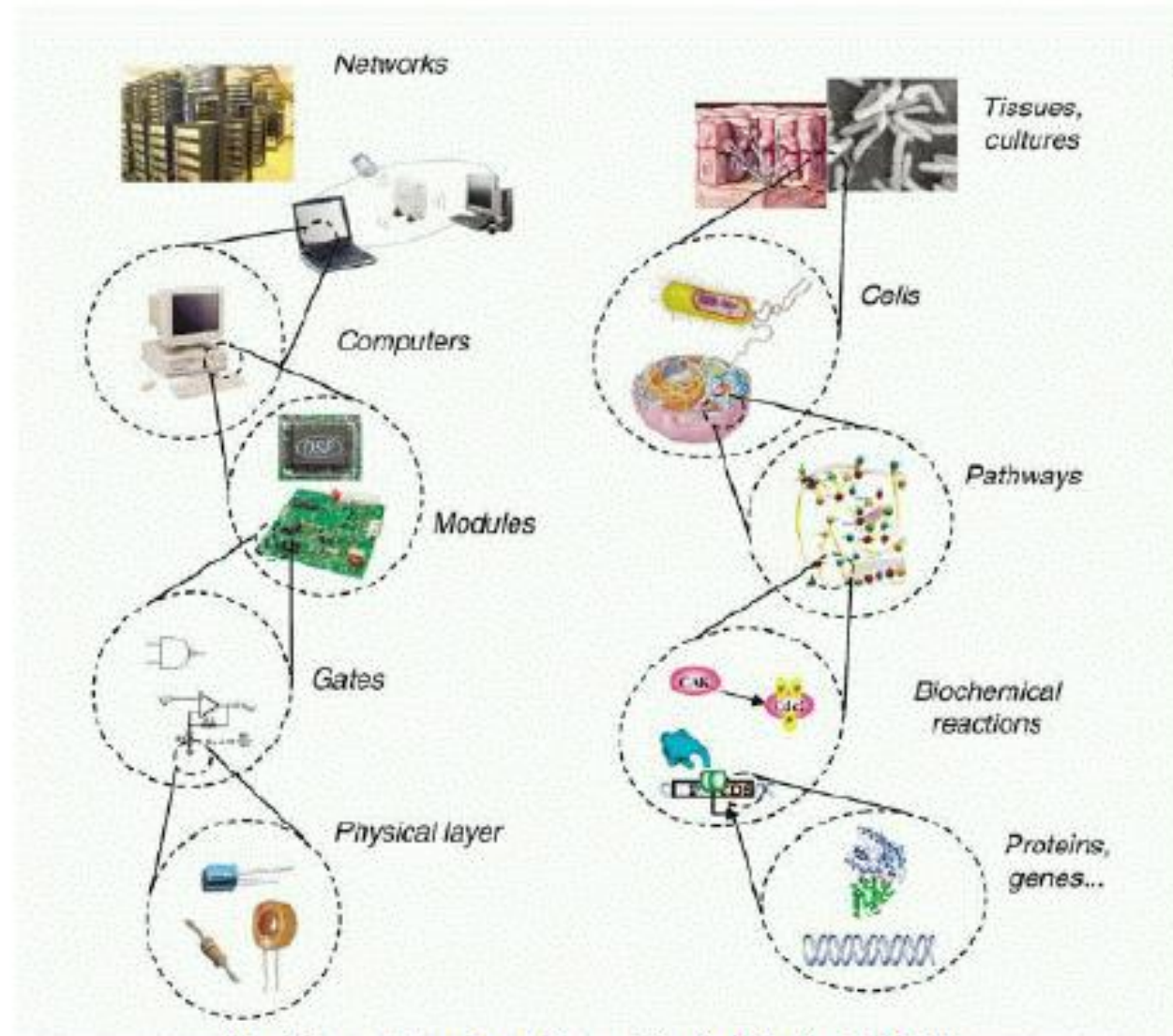
Community = Different populations (Biotic Factor = living)

Ecosystem = Various populations along with abiotic factors (non-living) coexisting.

Biome = Many ecosystems (Tundra, Tropical Rain Forest, Desert, etc...)

Biosphere = Many biomes

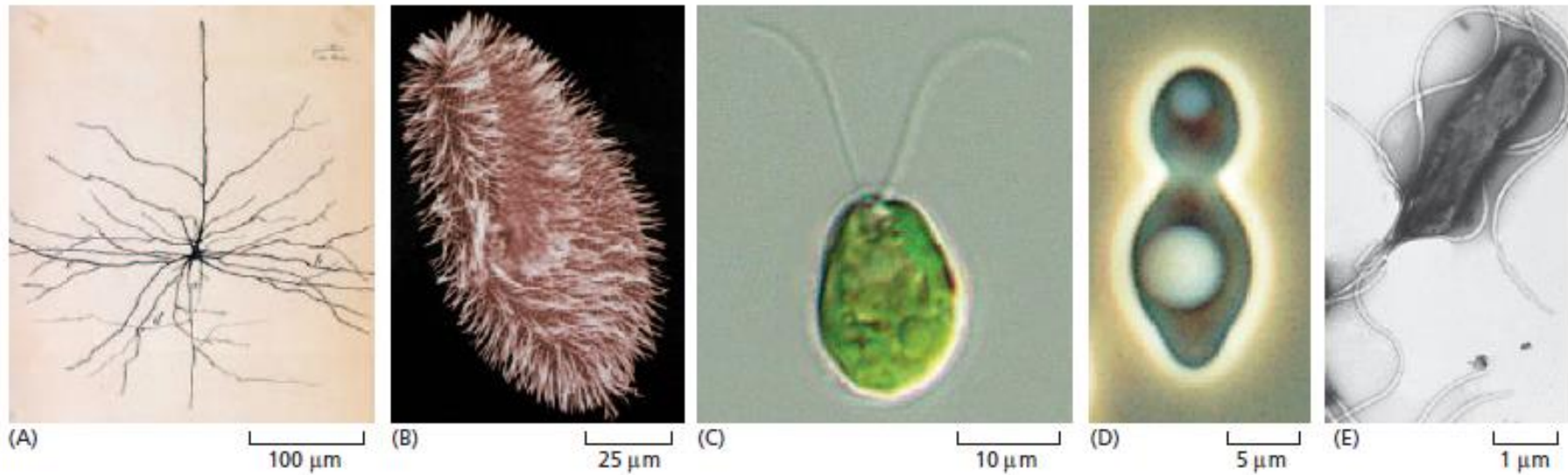
Cells compute?



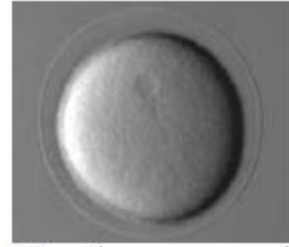
(Andrianantoandro, Basu, Karig, Weiss (2006))

Can we decipher the biological hardware and software?

Cells come in variety of shapes and sizes



Origin from single cell



(A)

100 μm



(B)

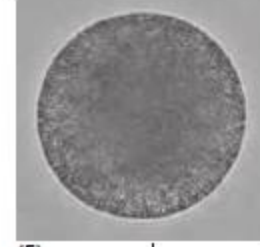


(C)

50 μm



(D)



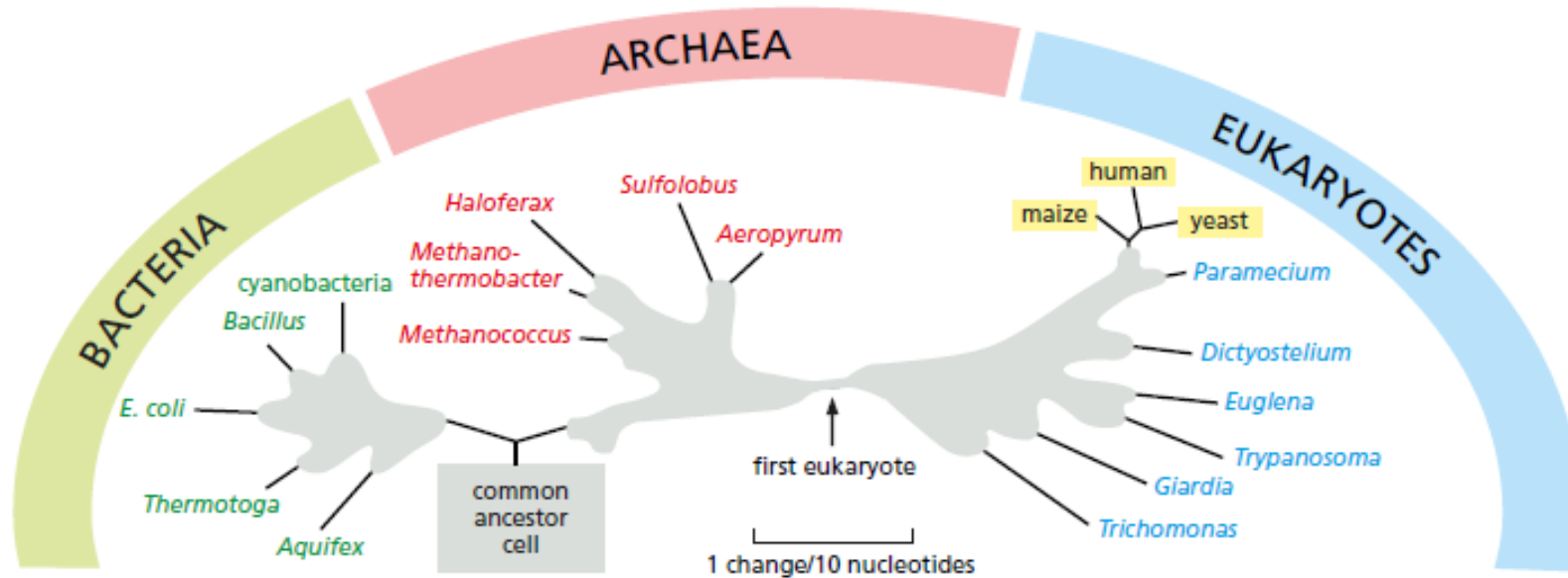
(E)

50 μm



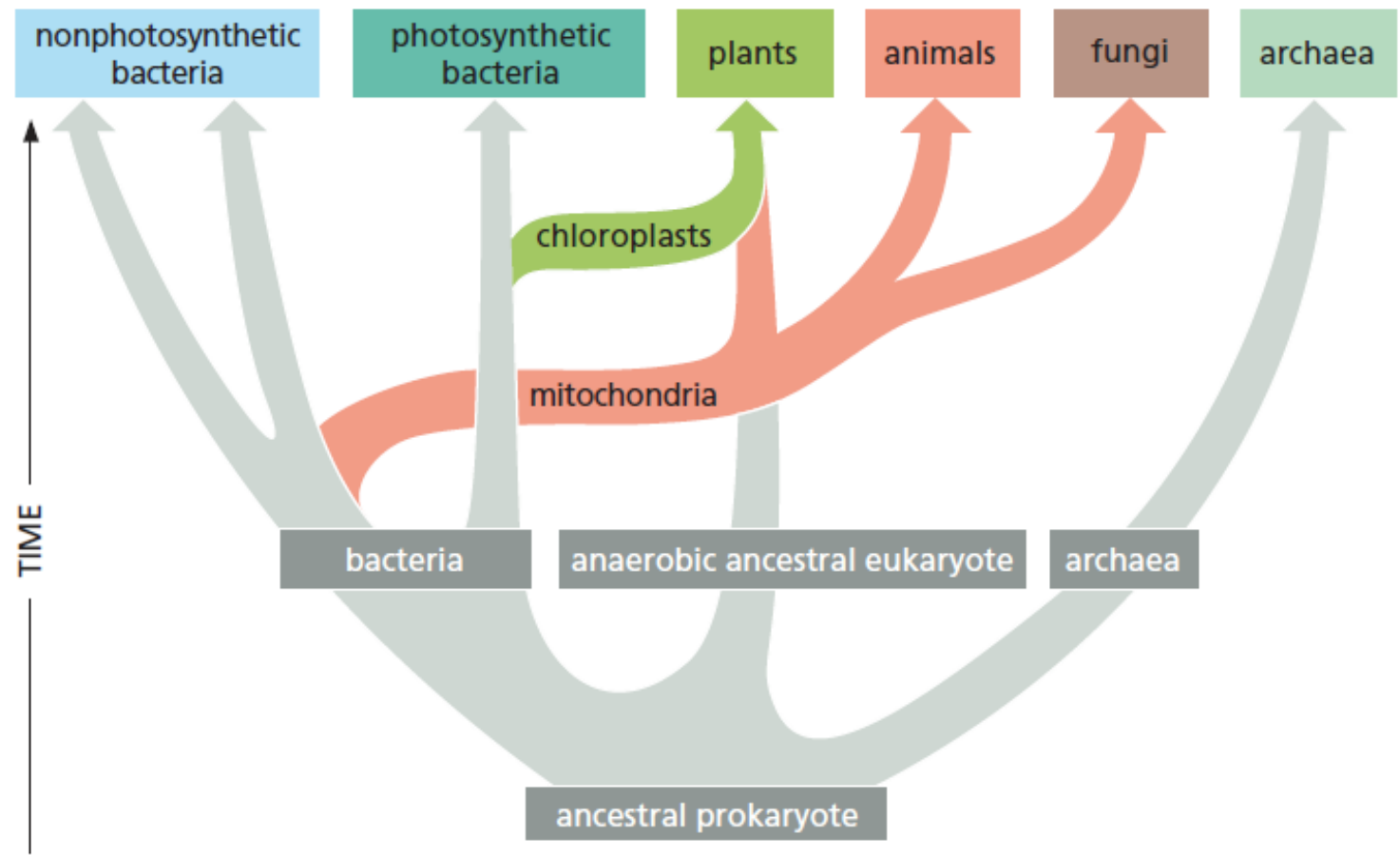
(F)

The Tree of Life

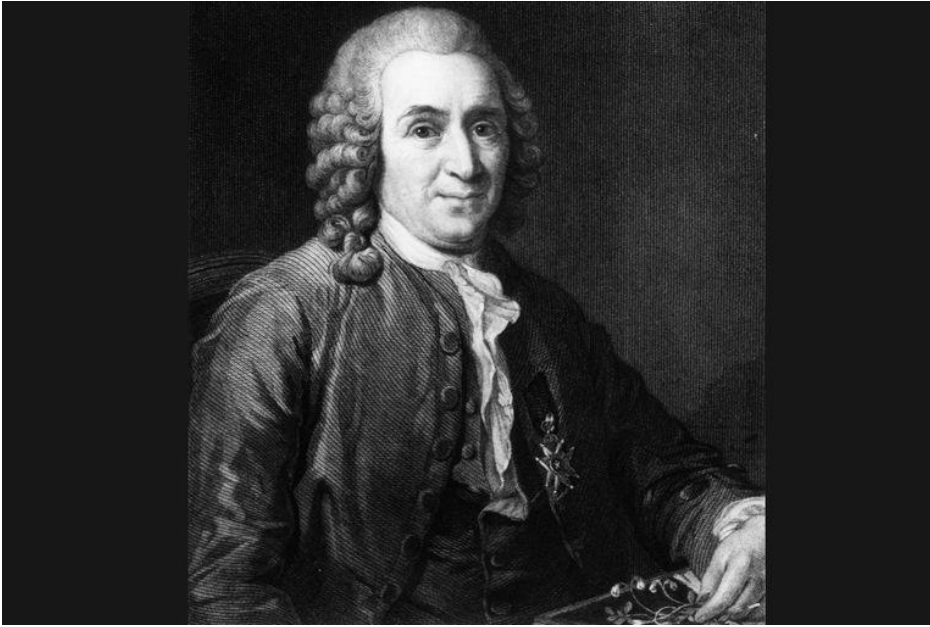


The domain system of classification was developed by Carl Woese and places organisms under three domains: **Archaea**, **Bacteria**, and **Eukarya**.

Under the domain system, organisms are further grouped into six Kingdoms. The Kingdoms include: **Archaeobacteria** (ancient bacteria), **Eubacteria** (true bacteria), **Protista**, **Fungi**, **Plantae**, and **Animalia**.



Taxonomy and Organism Classification



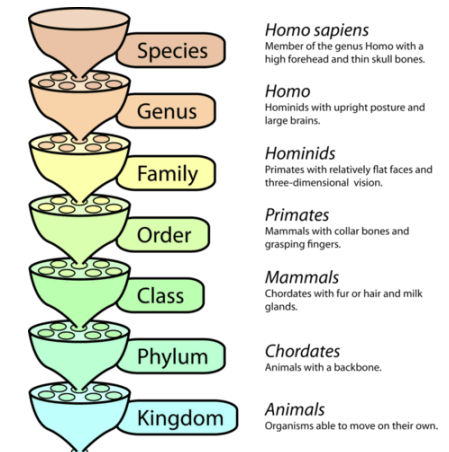
Botanist Carl von Linnaeus (1707-1778),
founder of the modern system of binomial
nomenclature for plants

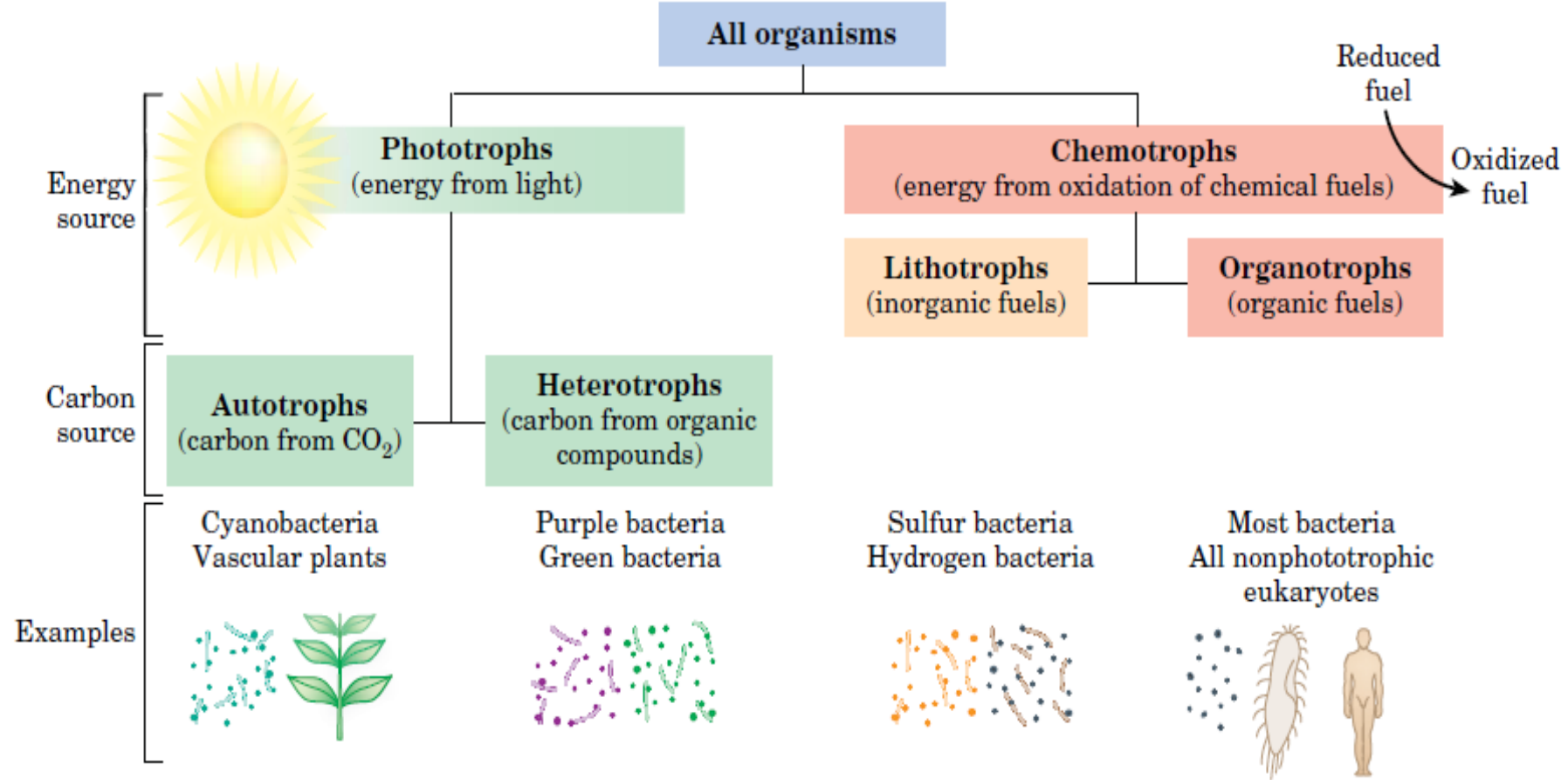
**Kingdom, Phylum, Class, Order, Family,
Genus, and Species.**

A helpful aid for remembering the
taxonomic categories:

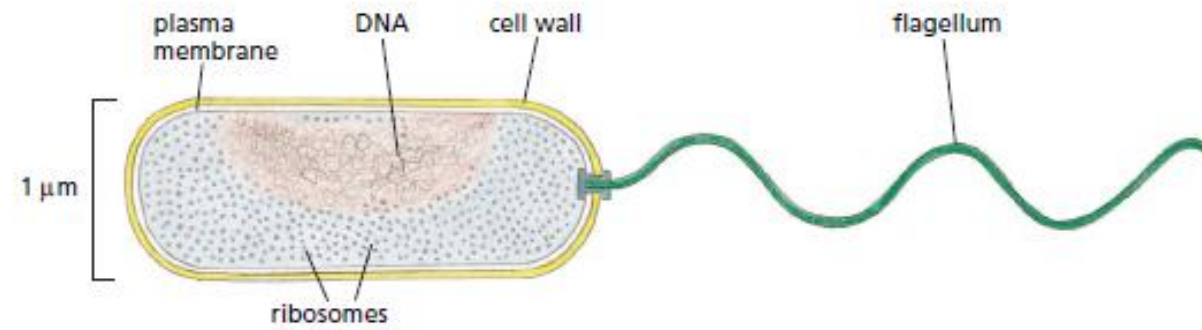
**Domain, Kingdom, Phylum, Class, Order,
Family, Genus, and Species** is the
mnemonic device:

Do Keep Plates Clean Or Family Gets Sick.

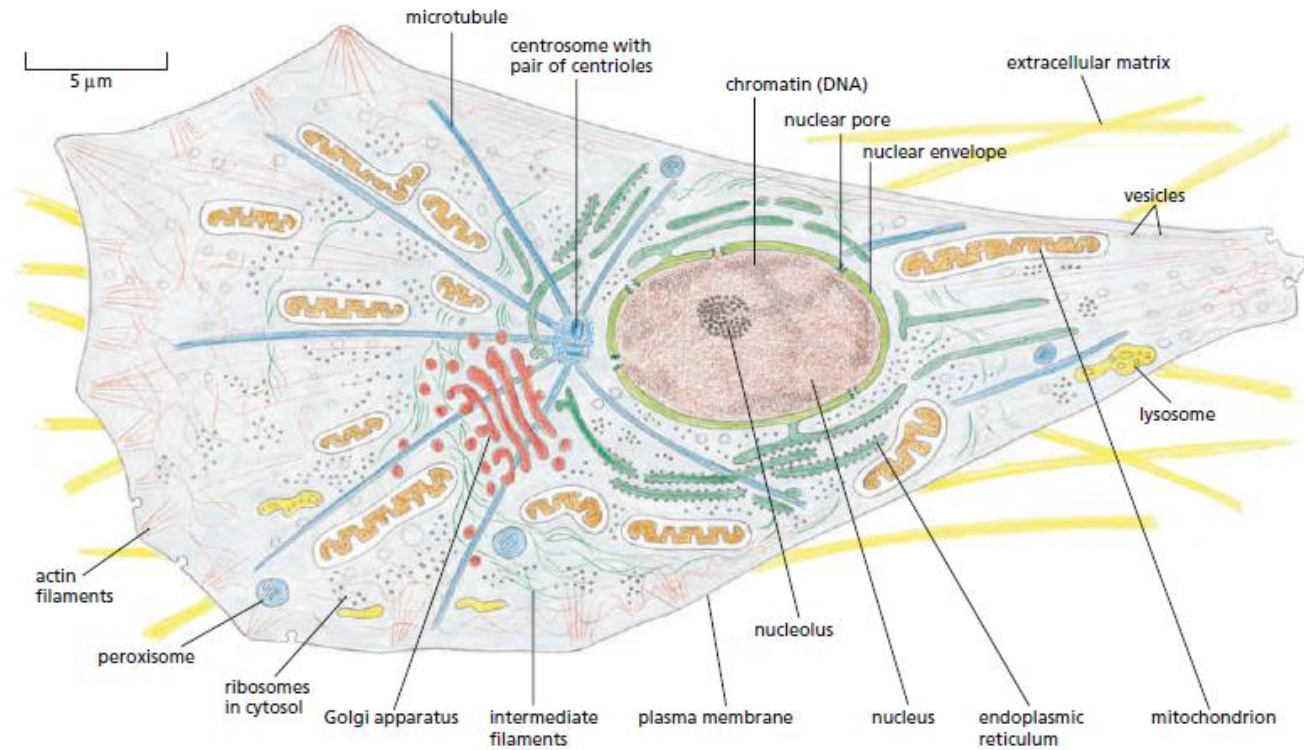




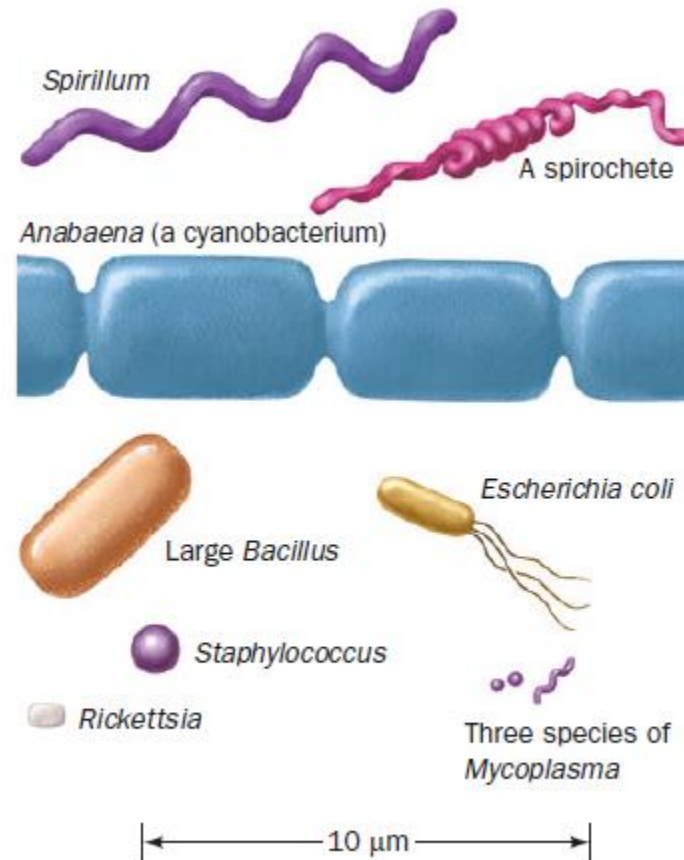
Bacterial cell



Animal cell



Scale drawings of some prokaryotic cells

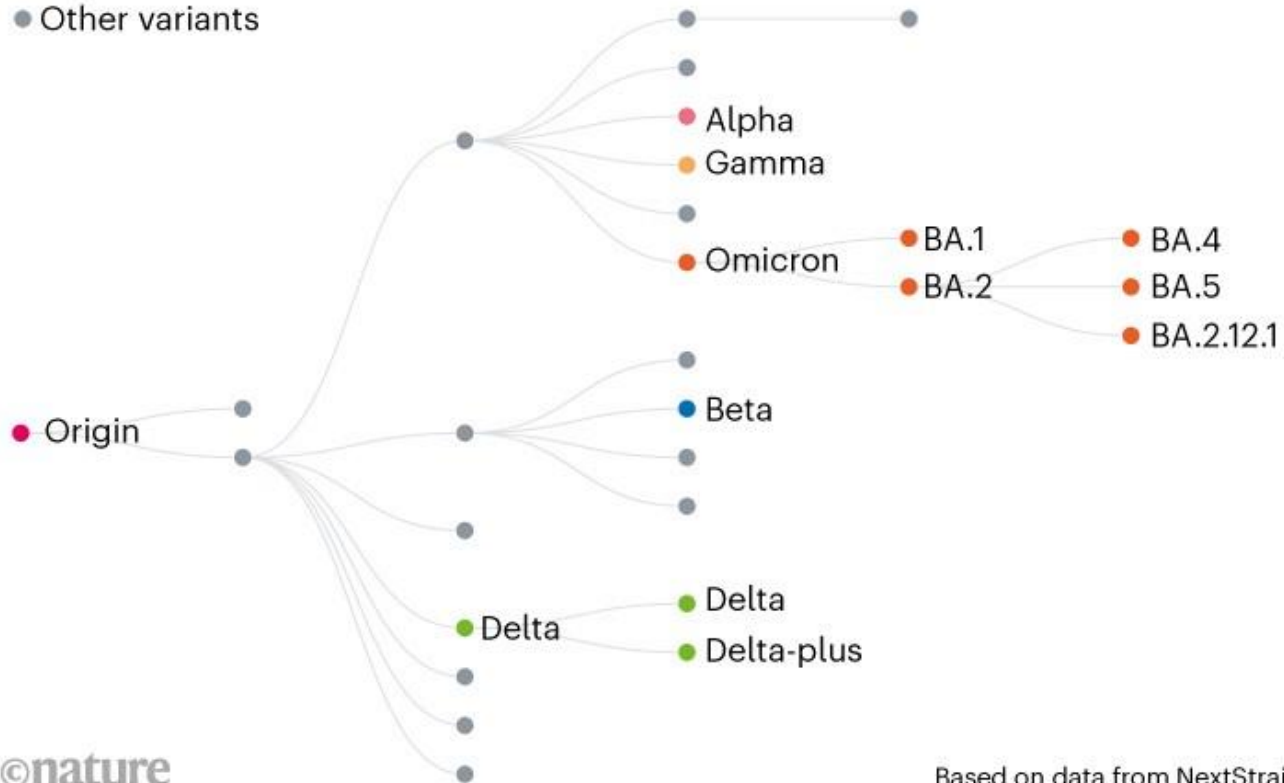


Three basic structures: spheroidal (cocci), rodlike (bacilli), and helically coiled (spirilla)

PATHOGEN PROGRESSION

This diagram shows how the coronavirus SARS-CoV-2 has evolved to spawn several related variants. The latest are BA.4 and BA.5 along the Omicron lineage, which has dominated infections this year.

● Other variants



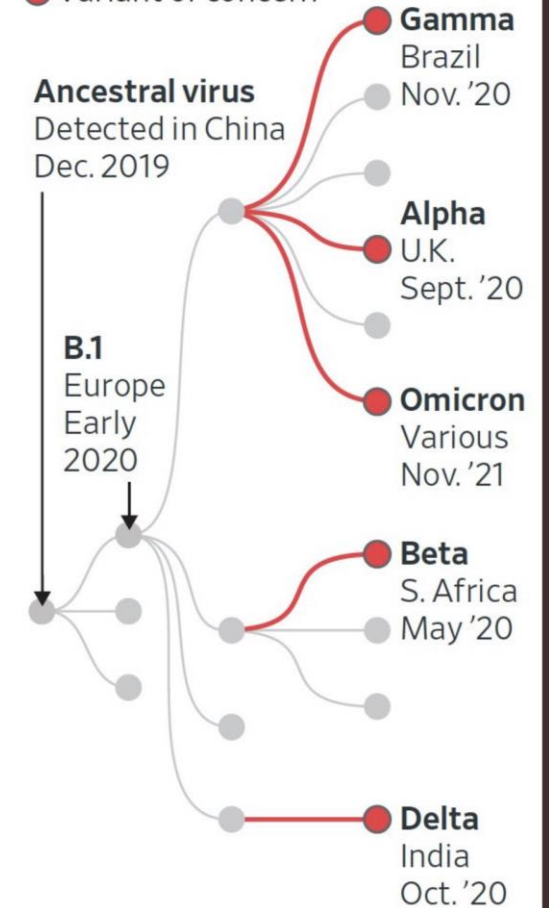
©nature

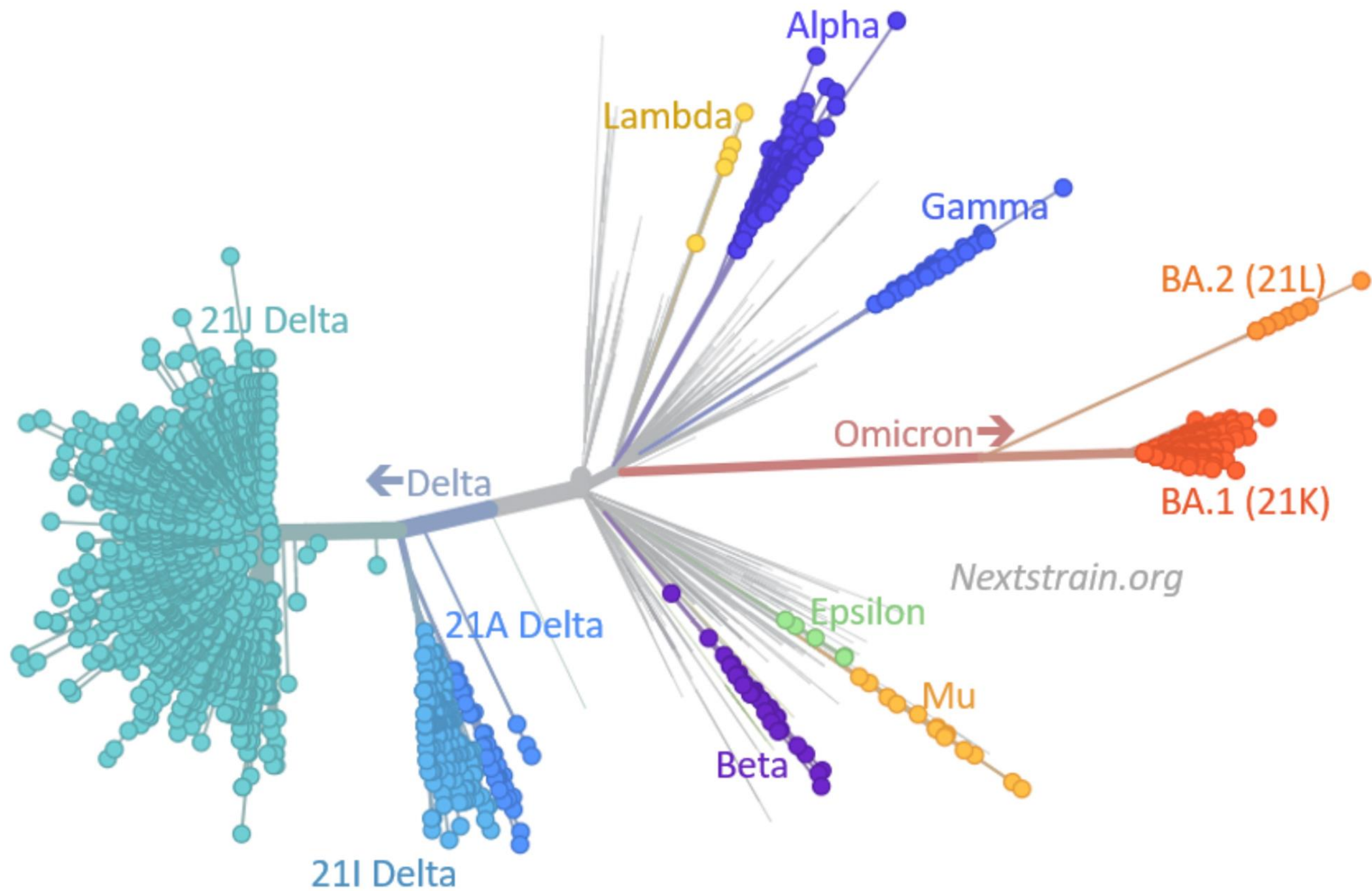
Based on data from NextStrain.

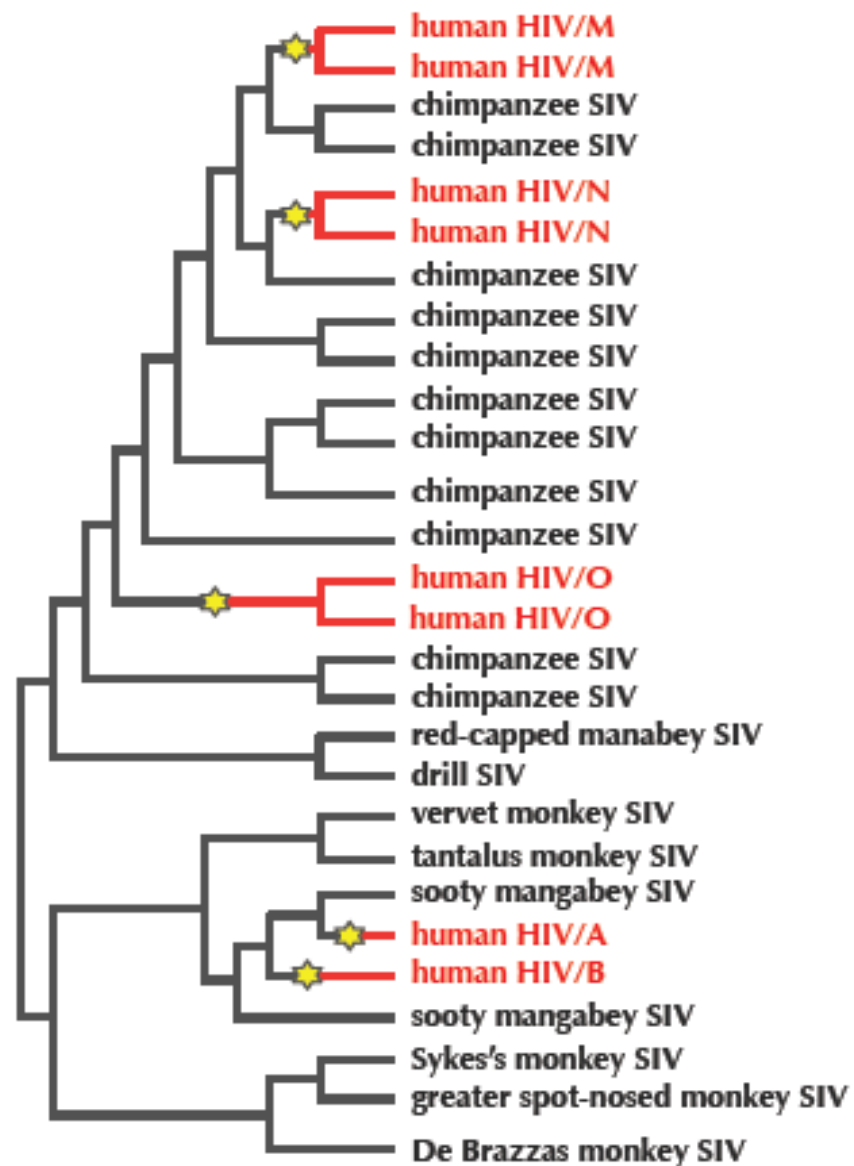
Evolutionary changes in the Covid-19 virus

● Variant of concern

Ancestral virus
Detected in China
Dec. 2019







Beta Coronavirus



Alpha Coronavirus

Let us play the sequence alignment game!

Strings ATGCATGC and TGCATGCA !

ATGCATGC
TGCATGCA

A**TGCATGC**-
-**TGCATGCA**

*We postulate a notion of a good alignment
as one that matches
as many symbols as possible.*

Strings ATGCTTA and TGCATTAA

A**TGC**-**TTA**-
-**TGC**A**TTA**A

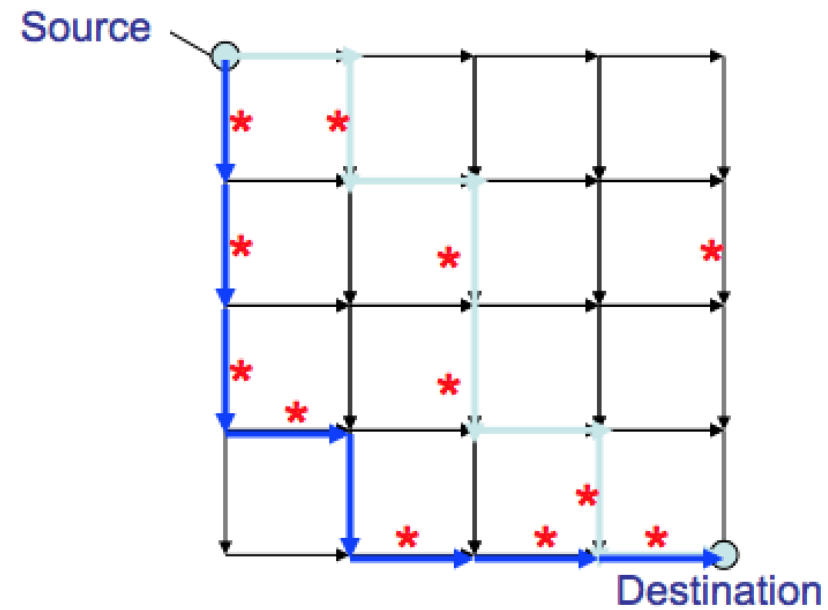
Longest Common Subsequence Problem:

Find a longest common subsequence of two strings.

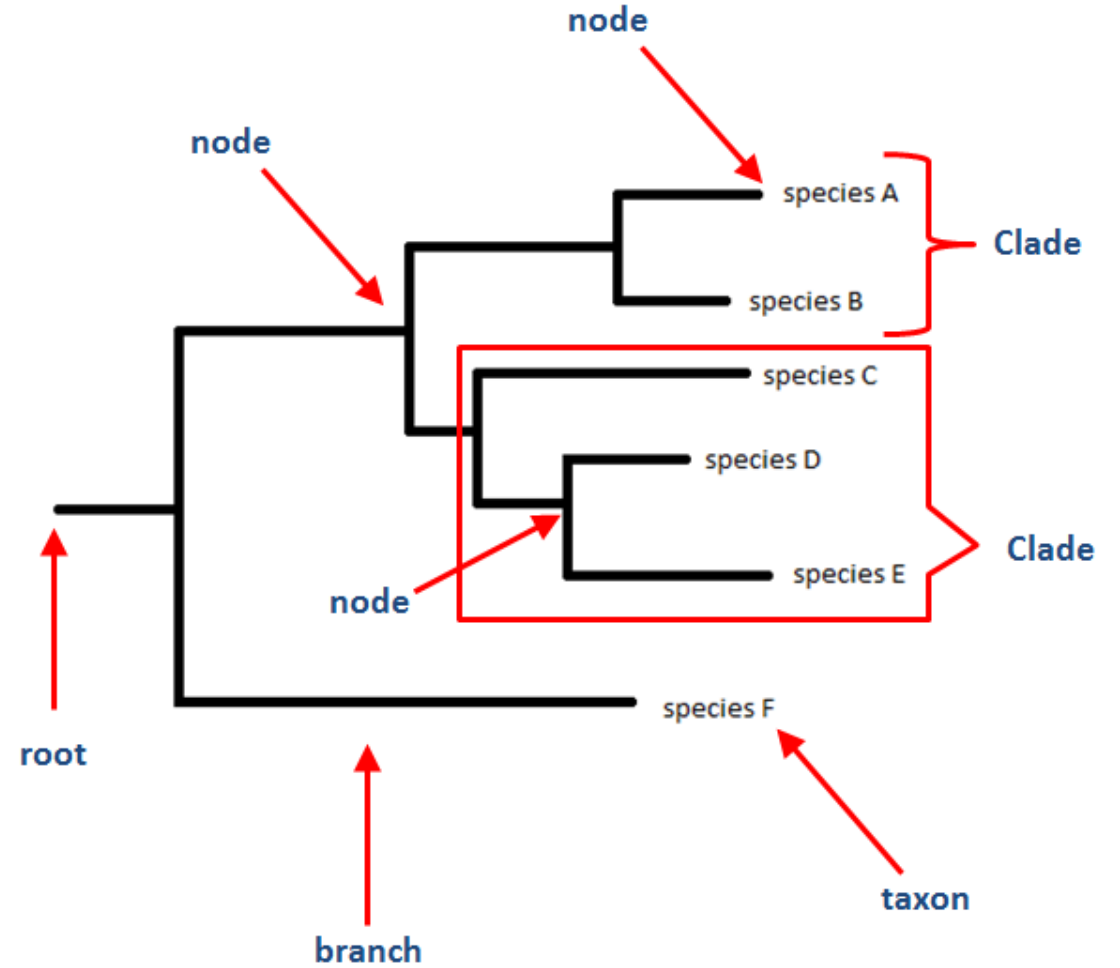
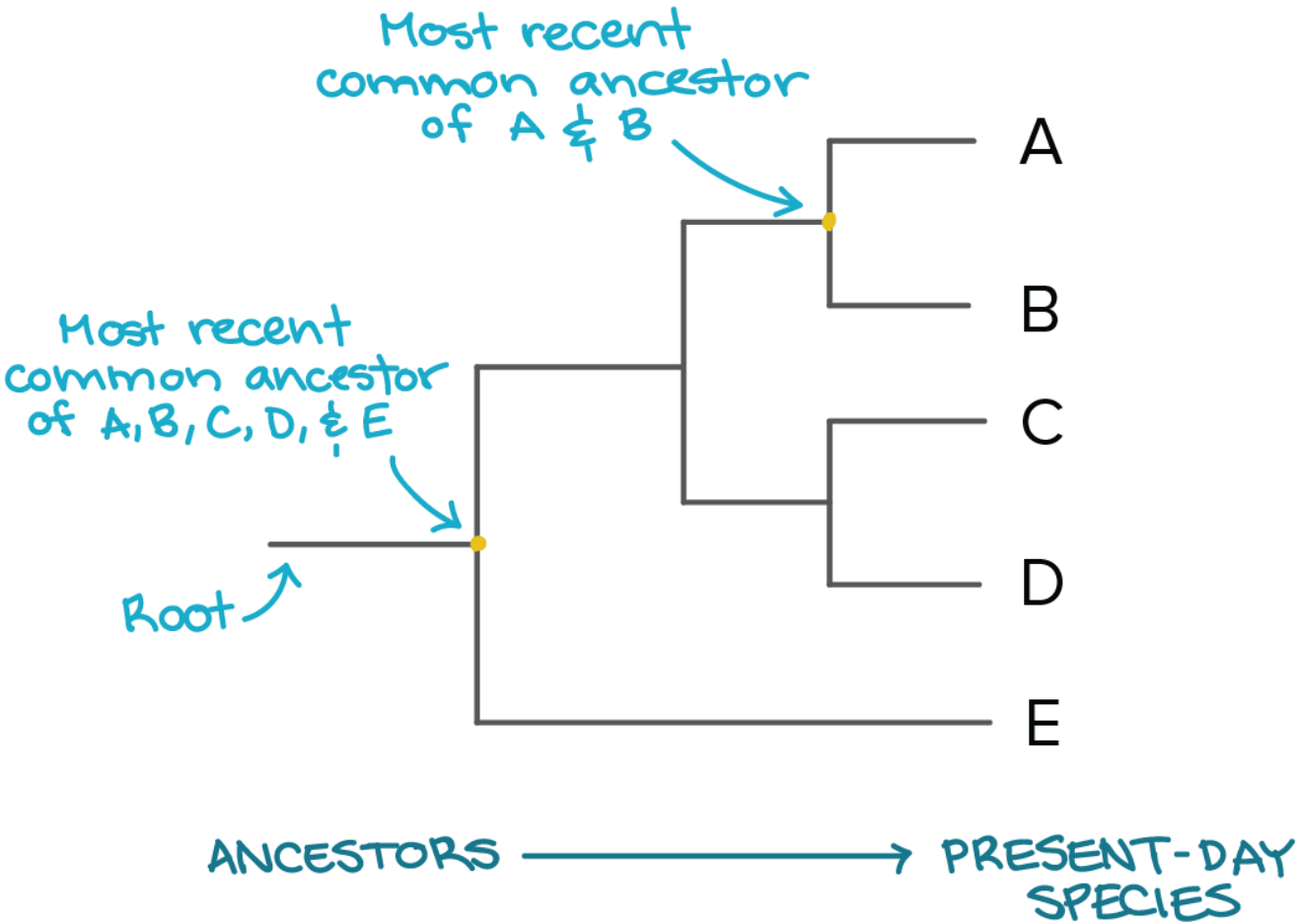
Input: Two strings.

Output: A longest common subsequence of these strings.

The Manhattan tourist problem!

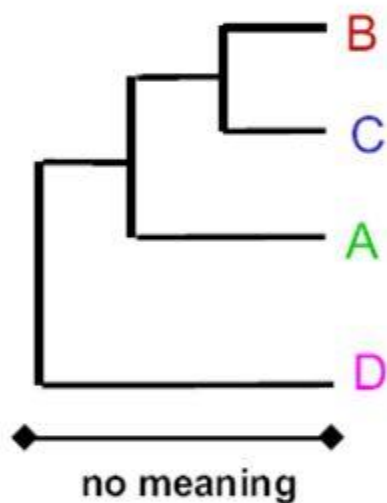


Parts of a phylogenetic tree



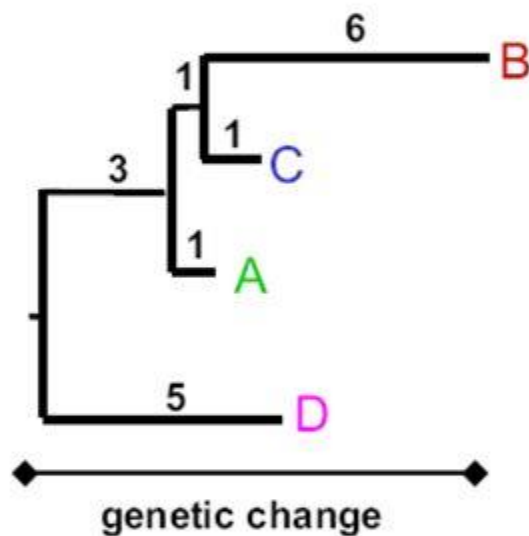
Types of trees

Cladogram



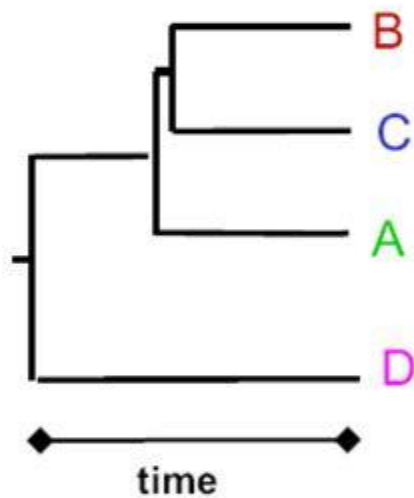
Simply shows relative recency of common ancestor

Phylogram

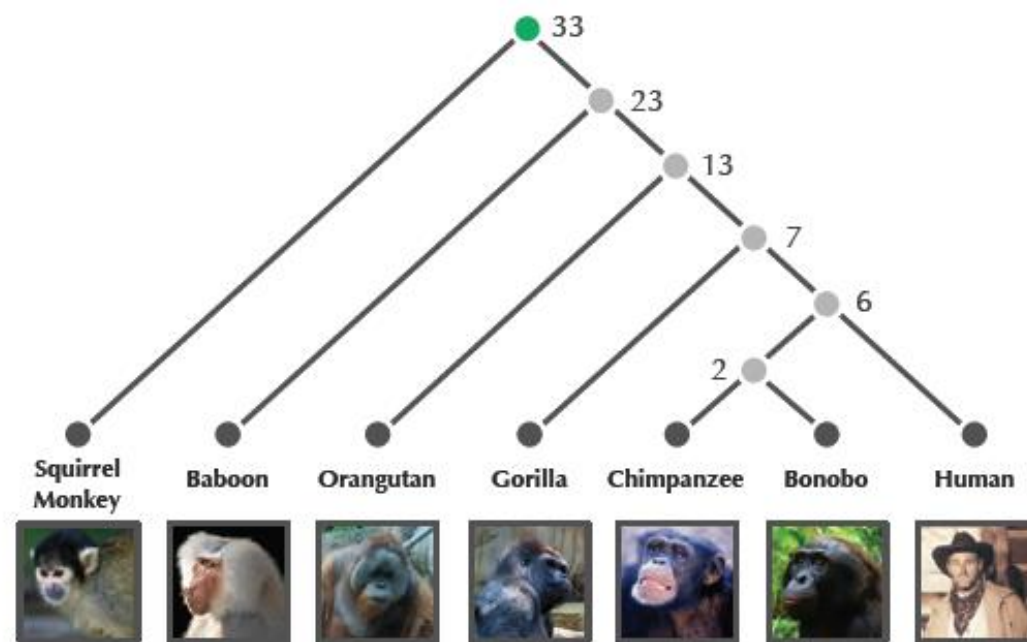
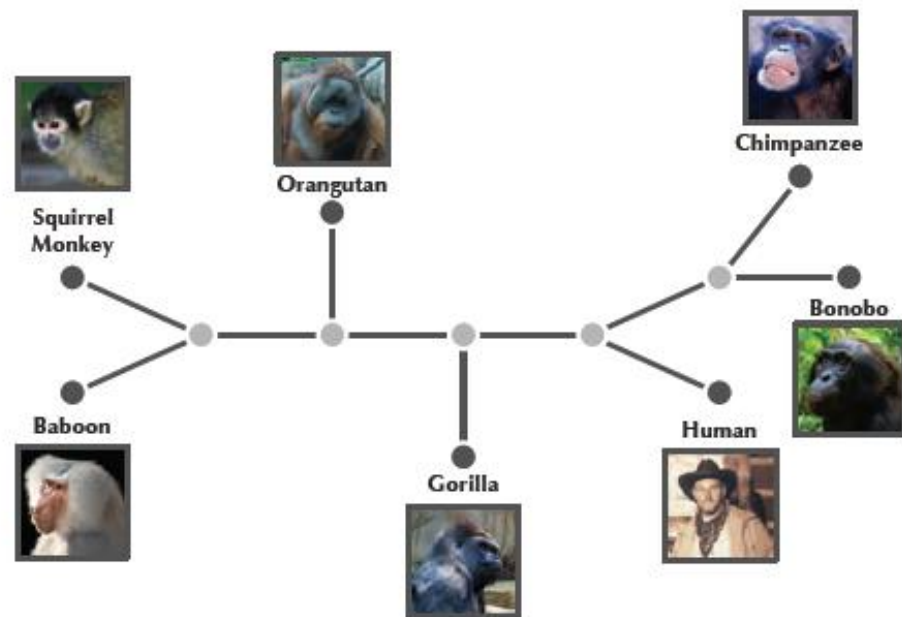
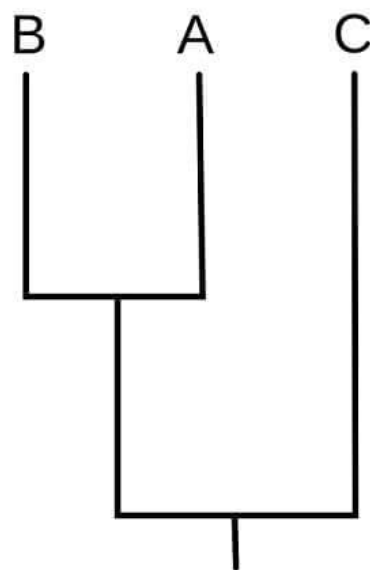
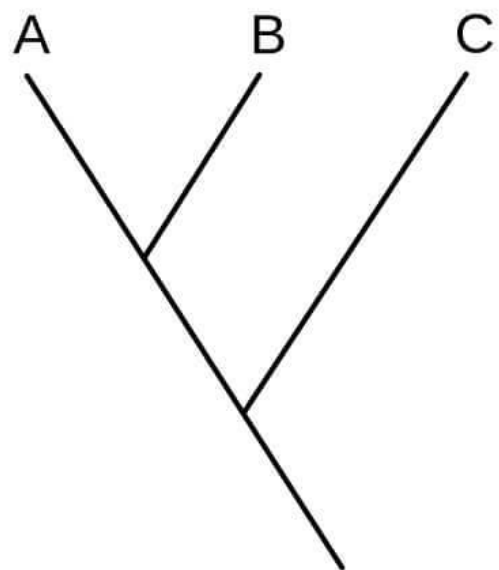


A cladogram with branch lengths

Ultrametric tree



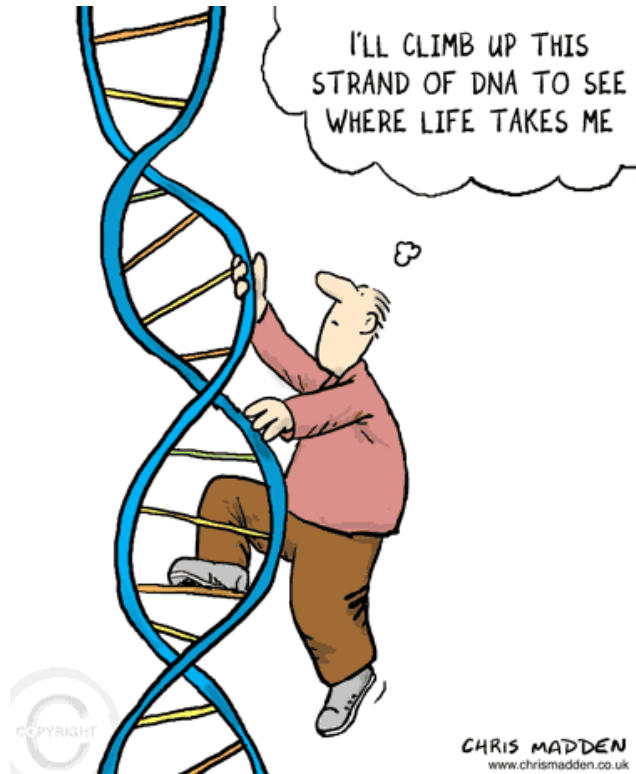
A dendrogram having all tips equidistant from root



Some definitions

- Classification - Grouping of things into classes.
 - e.g. Kingdom, phylum, class, order, family, genus, species.
- Taxonomy - Giving names to things.
 - e.g. Taxons -Mammalia, *Homo sapiens*, *Felis*
- Systematics- Understanding the relationships between things.
 - e.g. The daughter of my aunt is my cousin. The sister of my grandmother is my great-aunt.
- Phylogeny - The evolutionary history of a species.
 - If you don't know your history, you don't know anything!
- Cladistics - Phylogenetic systematics. Using the evolutionary history of species to understand their relationships, classify them, and give them names!

Universal feature-Chemical code



Organism	Genome size* (nucleotide pairs)	Approximate number of genes
<i>Escherichia coli</i> (bacterium)	4.6×10^6	4300
<i>Saccharomyces cerevisiae</i> (yeast)	13×10^6	6600
<i>Caenorhabditis elegans</i> (roundworm)	130×10^6	21,000
<i>Arabidopsis thaliana</i> (plant)	220×10^6	29,000
<i>Drosophila melanogaster</i> (fruit fly)	200×10^6	15,000
<i>Danio rerio</i> (zebrafish)	1400×10^6	32,000
<i>Mus musculus</i> (mouse)	2800×10^6	30,000
<i>Homo sapiens</i> (human)	3200×10^6	30,000
*Genome size includes an estimate for the amount of highly repeated DNA sequence not in genome databases.		

